

Assignment #3 – Cartesian motion system
ME 360 – Electromechanical Design
Spring 2025, release date: 03/26/2025

This is a team assignment that will take 5 weeks to complete and is due in class on Wednesday 04/30 for sections A1 and A2, and Tuesday 04/29 for sections A3 and A4.

The goal of the assignment is to apply the concepts and methods that we have reviewed in class to the design of a motion system with 2.5 degrees of freedom (DOF). You will build the mechanical and electrical sub-systems and integrate them to demonstrate the functionality of your prototype.

The learning objectives are:

- To devise different ways to embody ideal joints such as revolute, prismatic, spherical and/or universal joints.
- To design mechanical components with rational shapes, considering their function, mechanical loading, required accuracy, method of fabrication and assembly.
- To integrate and test a product prototype with mechanical and electrical components.

Form teams of 1-4 members and submit the list using this form: [LINK](#)

Or copy and paste this in your browser's address bar:

<https://docs.google.com/forms/d/e/1FAIpQLSdghQpdPl8eHUQYvPBpZsrPhRijcczwN6q72VKW4FaSuDh5iQ/viewform>

DELIVERABLE 1– Physical prototype

You will design and build a 2.5 DOF system that performs a task of your choice. The work area of the end effector will be at least 2.5 in x 2.5 in. This is the space in which the end effector (extruder, gripper, etc.) will move, not the size of the whole system.

Your design challenge is to build a system that is small and light enough to be easily carried in a backpack. You may design the device to be totally or partially disassembled for transport, as long as it can be assembled by one person in no more than 10 minutes.

The device will comprise the structure and motion elements (stepper and servo motors, pulleys, linear guides and belts).

You will be provided these standard components: 25 mm x 25 mm or 12.5mm x 25 mm 8020 aluminum extrusion, stepper motors, and the required belts and pulleys (timing and idler) for up to three linear stages. The actuators for the end effectors will depend on the application chosen by your team, for example they could be a solenoid or a servomotor.

The aluminum extrusion will serve as both structural material and linear guide. This is to minimize machining operations. The stepper motors may also be smaller than the ones you know, depending on the requirements of your application.

Except for the standard components, you will design all parts for 3D printing or laser cutting. Additionally, standard fasteners such as screws, pins and others may be used, depending on the requirements of each design. The EPIC has a large assortment of screws.

Teams that propose innovative applications that require the design of a purpose-built end effector, and which construct a prototype of such part, will receive extra credit.

DELIVERABLE 2 – An entry in your design portfolio

Deliverables for the project will be due on the dates indicated in Table 1. Progress on each deliverable will be reviewed and assessed at your team's meeting on the same week.

Submit your deliverables using this form: [LINK](#)

Or copy and paste this in your browser's address bar:

<https://docs.google.com/forms/d/e/1FAIpQLScI18RxI6DvamoFGuPtuyqYSLNce9rcQdJXCVL-PRcjp9eL7g/viewform>

Table 1 – Schedule of deliverables

DUE	DELIVERABLES
1st meeting	Team meetings start. Work plan and distribution. Conceptual design sketches.
2nd meeting	Team meetings. First linear stage ready for testing.
3rd meeting	Team meetings. All linear stages and end effector ready for testing (mechanical subsystem). Complete prototype, excluding electronics.
-	COMPLETE ALL ELECTRONICS AND CONTROL BEFORE PROTOTYPE TESTING AND FINE-TUNING DAY
04/23 (A1, A2) 04/24 (A3, A4)	Prototype testing and fine-tuning
04/30 (A1, A2) 04/29 (A3, A4)	Project demonstrations. Prototype ready for testing in class.
05/02	An entry in your design portfolio.

Team meetings with the instructor

I will meet each team independently for 20 minutes EVERY WEEK to review your designs. ALL team members must attend the meetings and contribute to the design. You will receive a peer evaluation form in due course to keep track of individual participation. Outside of our meeting times, teams may use class time to work on their projects.

The meeting times will be used to track your progress and to support you in the different stages of the project. Meeting times are listed at the end of this document and will be conducted in room ENG 404. Come prepared to discuss your project. All electronic files should be stored in a drive accessible to all team members.

The time slot for your team's meetings with the instructor will be posted once the teams are formed.

All team members must participate in the design, construction, decision-making, troubleshooting, presentation, and demonstration of the prototype.

Grading rubric

- 1) To what degree does the device perform the expected function?
- 2) To what degree is this an original idea/application? (Not a reproduction of existing systems).
- 3) Does the design meet the requirements indicated on the assignment sheet? (Dimensions, ease of assembly/disassembly).
- 4) Did the team design an original end effector?
- 5) Were all joints correctly designed for assembly and disassembly? (Do not use tape or glue, design components with the appropriate fixtures for their assembly: snap fits, screws, etc.).
- 6) Safety and craftsmanship (Do not leave sharp edges, make sure no parts come off and no part of the prototype needs to be held by hand during operation).
- 7) Economy of the design (simple, effective use of materials and components).
- 8) Robustness and reliability (Structurally and electrically robust, tolerant to the classroom environment, works reliably).